

Bandit REINFORCE Hyperparameter Sweep Report

1. Algorithm Overview

The core algorithm used in this experiment is **Bandit REINFORCE**, an independent policy gradient model utilizing a PyTorch Multi-Layer Perceptron (MLP). It replaces deterministic action-bound algorithms (like UCB) with a differentiable, stochastic deep learning approach to optimize multi-agent traffic routing.

1.1 Action Selection

The agent's neural network outputs a vector of scores (logits) for each available route. These logits are scaled by a **Temperature** (τ) parameter to control the sharpness of the distribution, and are then converted into probabilities using a Softmax function:

$$\pi(a|s) = \frac{\exp(Z_a/\tau)}{\sum_i \exp(Z_i/\tau)}$$

(Where Z represents the logits output by the neural network for action a)

1.2 Policy Update (REINFORCE)

Once the vehicle completes its route, it receives a reward R (negative travel time). The algorithm uses a baseline b to reduce variance. The weights of the neural network θ are updated using gradient ascent, scaled by the **Learning Rate** (α):

$$\theta \leftarrow \theta + \alpha \nabla_{\theta} \log \pi(a|s) (R - b)$$

1.3 Exploration via Entropy

To prevent the agent from permanently committing to sub-optimal routes early in the training phase, an **Entropy Bonus** is added to the objective function, scaled by the **Entropy Coefficient** (β):

$$H(\pi) = - \sum_a \pi(a|s) \log \pi(a|s)$$

$$\text{Objective} = \text{Expected Reward} + \beta H(\pi)$$

2. Experimental Setup

The hyperparameter sweep consisted of **54 total parallel jobs**, evaluating every combination of the following parameters across **3 random seeds**:

- **Learning Rate (α):** `3e-4` , `1e-3`
- **Entropy Coefficient (β):** `0.01` , `0.05` , `0.10`
- **Temperature (τ):** `0.5` , `1.0` , `2.0`
- **Random Seeds:** `42` , `123` , `7`

Each job executed 500 training episodes and 100 testing episodes on the `ingolstadt_custom` network map.

3. Prerequisite Check: The TraCI Handle Bug

Context

Another prerequisite flagged in the documentation questioned whether `env.simulator.sumo` was the correct handle to access the active SUMO simulation. If incorrect, the connection would fail and the algorithm would fall back to feeding the neural network neutral constants `[1.0, 0.0]` instead of real-time traffic data.

Investigation & Resolution

To verify this, the internal source code of the RouteRL simulator (`/home/sathyakumarnandakumar/URB/urbenv-github/lib/python3.12/site-packages/routerl/environment/simulator.py`) was analyzed. It was discovered that the true active handle was `env.simulator.sumo_connection`. The Bandit REINFORCE script was patched prior to the sweep to use this correct handle, ensuring the neural network received actual, real-time congestion data throughout the entire 54-job execution.

4. Results and Rankings

The results demonstrate the stability of the Bandit REINFORCE algorithm in this topology. The average travel time for Autonomous Vehicles (CAVs) across all configurations fell into a tightly clustered band between **4.328 seconds** and **4.403 seconds**.

Top 5 Configurations

Ranked by the lowest average CAV travel time across all 3 seeds.

Rank	Learning Rate	Entropy	Temperature	Average CAV Travel Time (s)	Overall Test Travel Time (s)
1	1e-3	0.05	0.5	4.328	4.282
2	1e-3	0.10	1.0	4.363	4.294
3	1e-3	0.10	0.5	4.365	4.292
4	3e-4	0.01	0.5	4.373	4.293
5	1e-3	0.01	1.0	4.374	4.298



Bottom 5 Configurations

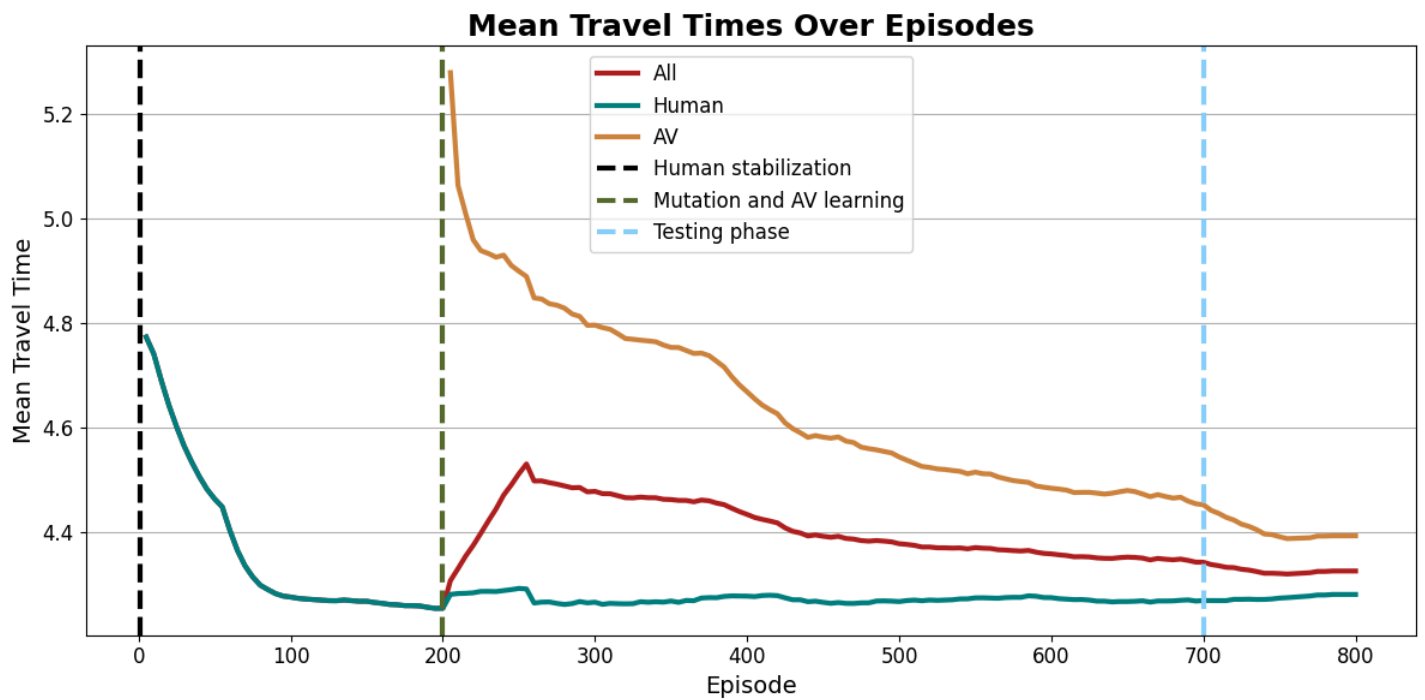
Rank	Learning Rate	Entropy	Temperature	Average CAV Travel Time (s)	Overall Test Travel Time (s)
50	3e-4	0.10	1.0	4.393	4.306
51	3e-4	0.01	2.0	4.395	4.311
52	1e-3	0.01	2.0	4.397	4.313
53	3e-4	0.05	0.5	4.401	4.312
54	3e-4	0.01	1.0	4.403	4.309

5. Visual Analysis

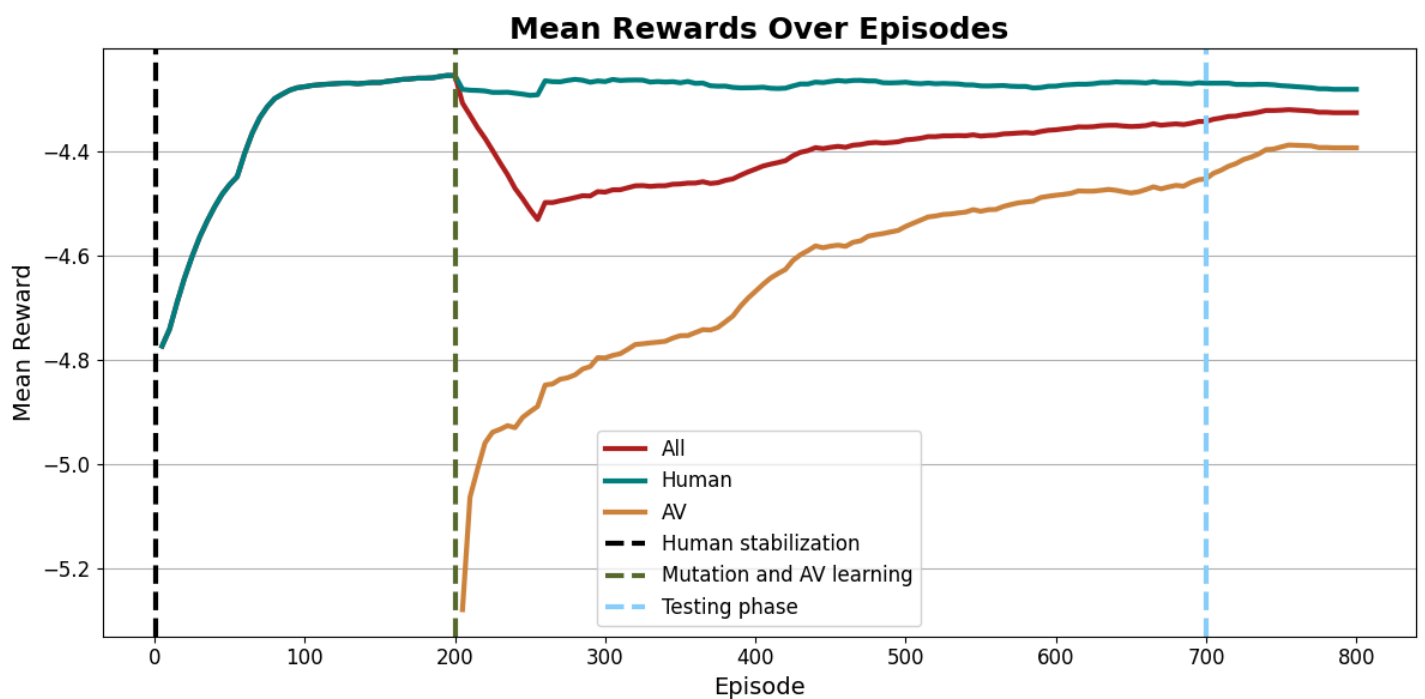
The Best Configuration

Parameters: `lr=1e-3` , `entropy=0.05` , `temperature=0.5` Notice the tight, highly stable convergence band during the testing phase.

Travel Times:



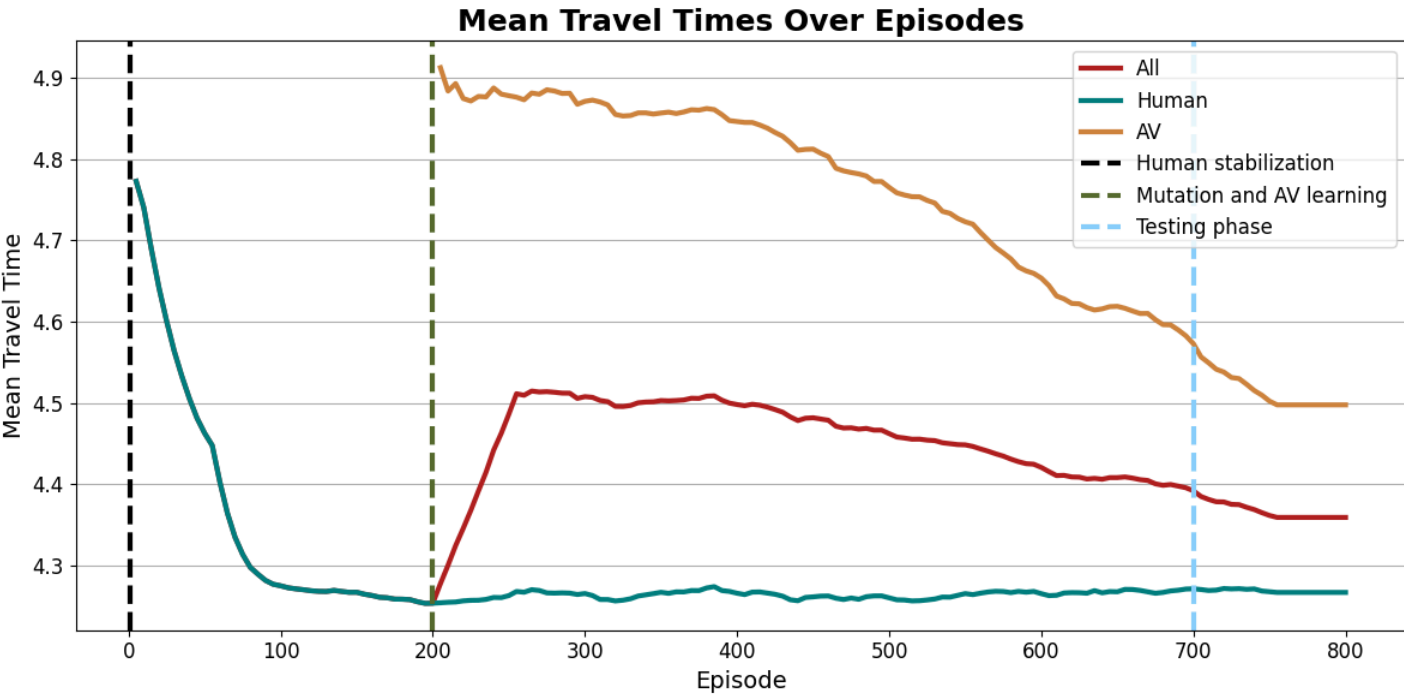
Rewards:



The Worst Configuration

Parameters: `lr=3e-4`, `entropy=0.01`, `temperature=1.0` While the convergence still succeeds, it is noticeably noisier with wider variance during both training and testing.

Travel Times:



Rewards:

